

Krishnaa Vadivel

Scientific Computing | Machine Learning | Electrical Engineering

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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Summary

Electrical-engineering PhD researcher who bridges computational materials physics, GPU/HPC scientific software, scientific ML, and embedded engineering. Select this broad version only when a role genuinely crosses those domains.

Technical Skills

Scientific compute: Python, C++, CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), MPI, OpenMP, PETSc, SLEPc, DOLFINx, JAX, PyTorch Geometric

Systems: Frontier, Perlmutter, Docker, Kubernetes, Linux, CMake, ASP.NET, Microsoft SQL Server

Hardware: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, board debugging, OpenGL

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Translate many-body and first-principles physics into distributed Python/C++ workflows using MPI, OpenMP, PETSc/SLEPc, CUDA/ROCm, and national-lab GPU platforms.
- Build DeePH-style computational graphs and scientific-ML models with equivariant GNNs and automatic differentiation for LiF and double-perovskite optical-property calculations.
- Combine simulation research with semiconductor and photonics lab experience; publish and present self-trapped-exciton research.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Shipped CUDA/OpenCL and OpenGL applications for acoustic-waveform processing and diagnostic visualization.
- Delivered C#, ASP.NET, and SQL Server systems for remote engineering-tool testing and sensor-calibration data.
- Implemented firmware/FPGA functionality and supported board-level integration, debugging, and field validation.

Selected Projects

Materials ML — Equivariant GNN and automatic-differentiation workflows for optical-property calculations.

Integrated photonics — Meep FDTD ring-resonator filter simulation, later fabricated and optically tested.

Finite elements — DOLFINx/PETSc heat, electrostatic, and flow simulations.

Education

PhD, Electrical Engineering

Yale University

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Research Record

Co-author, *Journal of the American Chemical Society*, 2025; APS presentations, 2024–2026.